Adelaide University

Business Management Systems

Processes and Automation

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We respectfully acknowledge the Kaurna, Boandik, and Barngarla First Nations Peoples and their Elders past and present, who are the Traditional Owners of the lands that are home to our campuses across Adelaide and South Australia.

Outline

- Projects vs. processes
- Example of processes
- Business process modelling
- Process streamlining and automation

Examples of Projects

- A company plans to start a new business
- A government group develops a system to track child immunizations.
- A department needs to respond to a legal challenge

Project Attributes

- A project has many attributes, including:
  - Has a unique purpose
  - Is temporary
  - Involves uncertainty

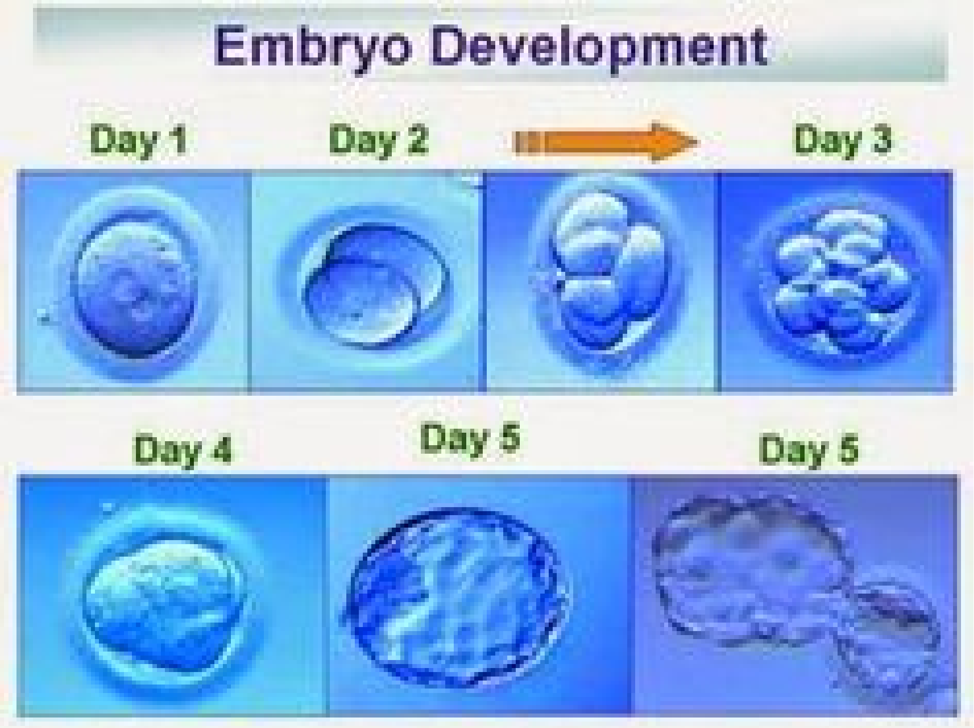
Project vs Process

- A project is a temporary endeavour undertaken to create a unique product, service or result.” (PMBOK p. 5)
- It is temporary in that every project has a definite starting and ending time. When the objectives of the project have been met the project ends.
- In contrast a process is part of day-to-day operation, a work that is continuously undertaken

Process	On-going, Repetitive, with renewable objectives Managed within the organisational structure
Project	Temporary, Unique objective, Carried across the organisation, with members from different groups

Example: In Vitro Fertilization

- What are the steps here?



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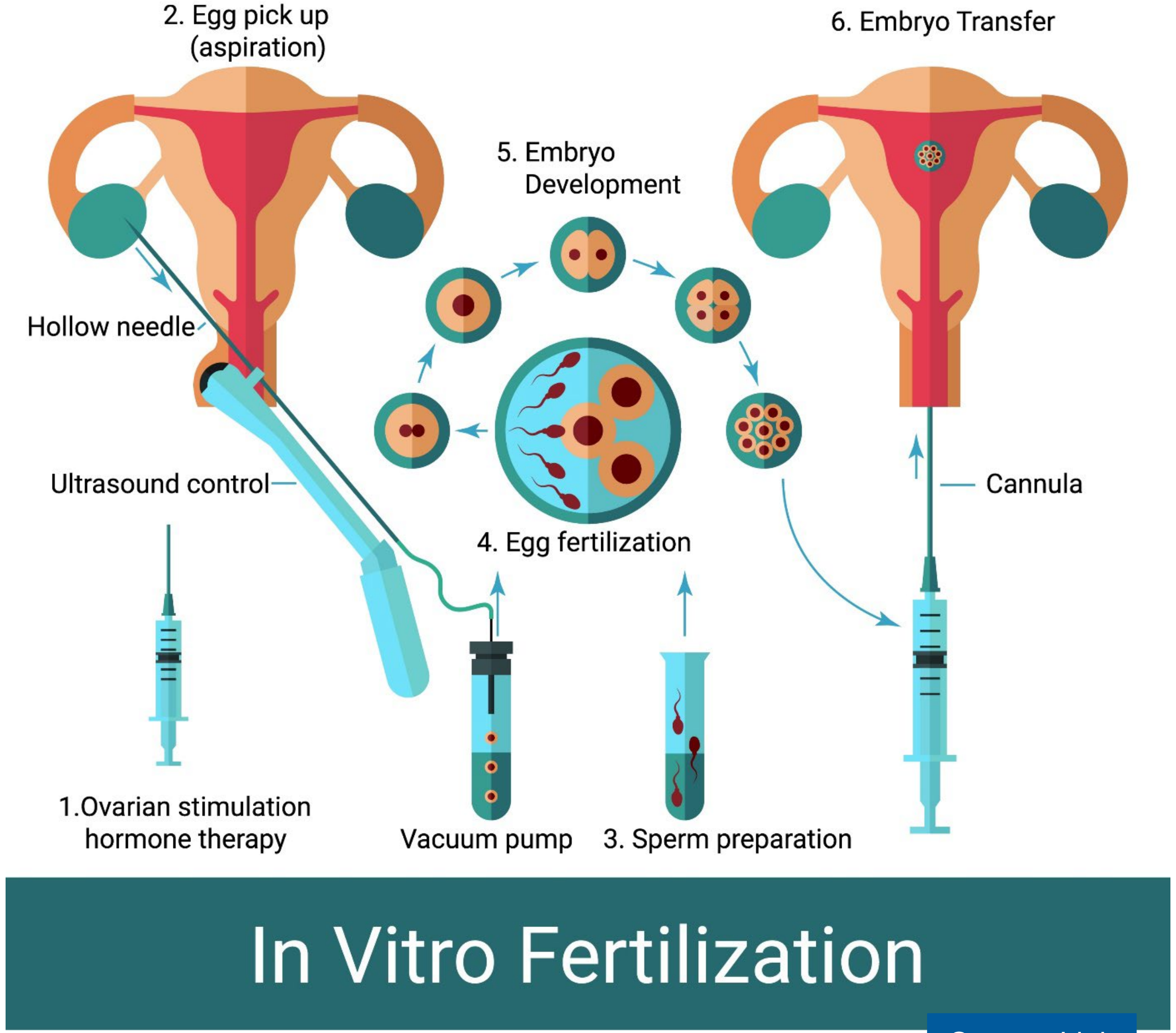


Source: Link

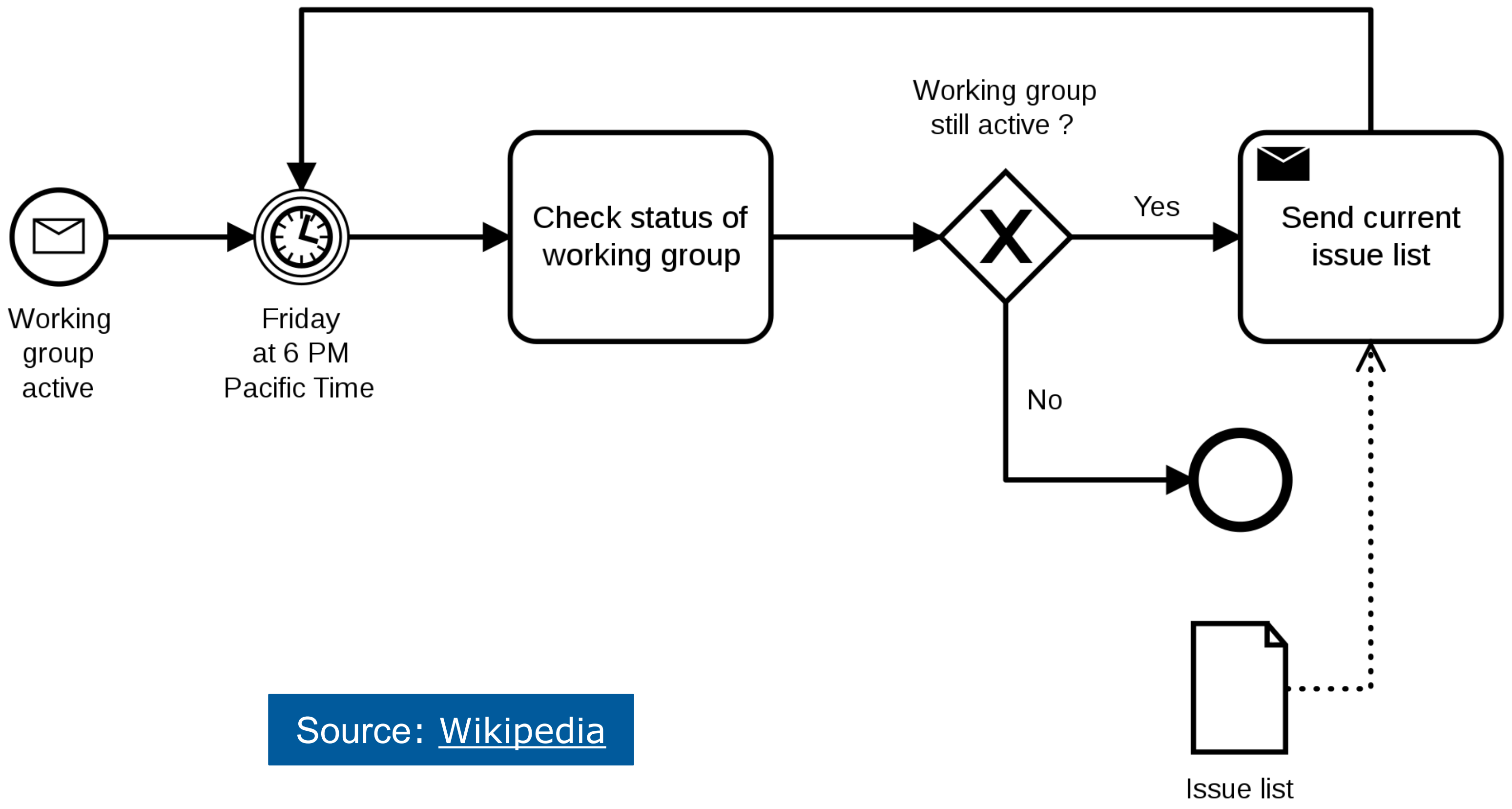
Example: Bank Loan Process

- A different process
- Banks may differ on how they go about approving a loan application

Source: [Link]

Business Process Modelling

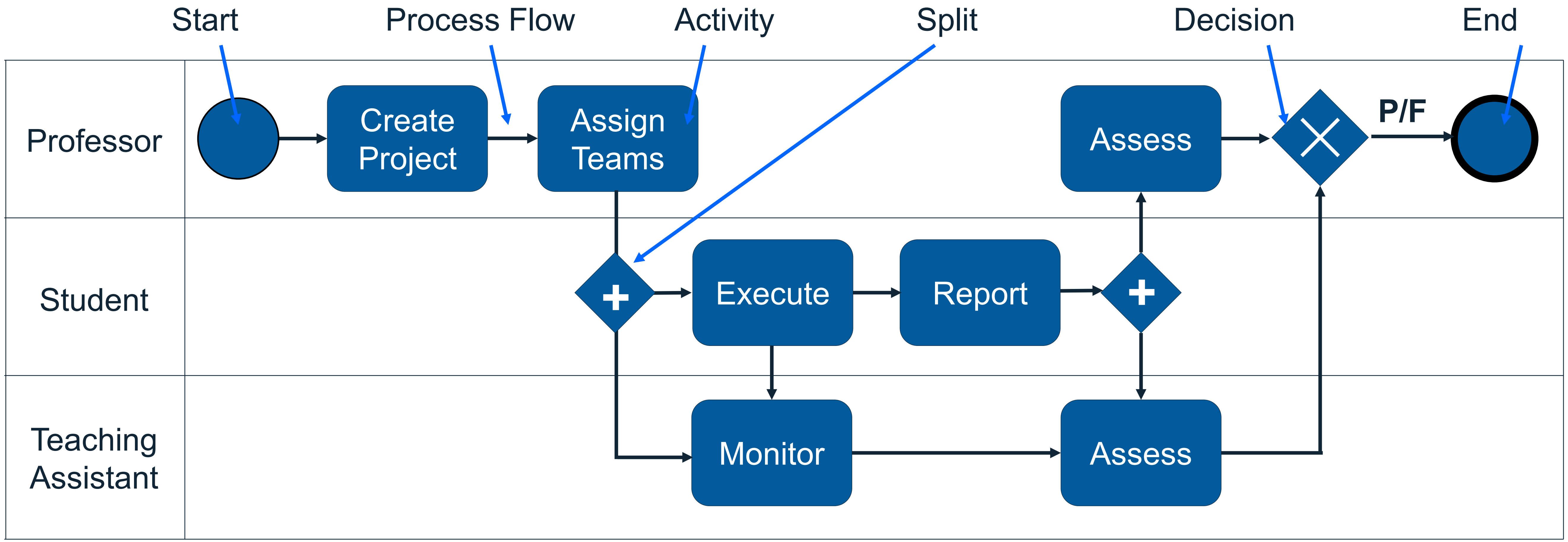
- Business process modelling (BPM) in business process management is the activity of representing processes of an enterprise
- The aim is to analyse the current process to improve and possibly automate
- BPM is typically performed by Business analysts, with modelling expertise
- Subject matter experts, with specialized knowledge of the processes



Source: [Wikipedia]

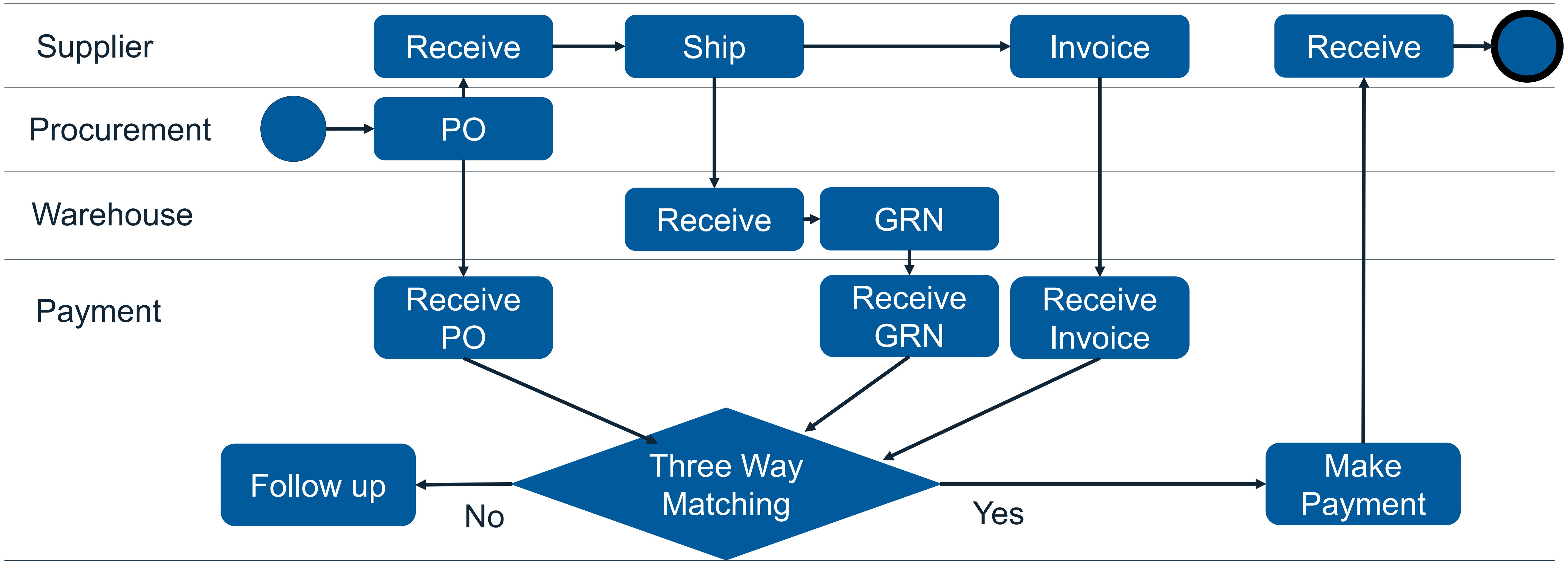
BPM Flow Charts and Notations

- A business process model may be drawn as a linear flow of events and activities. It is usually drawn as a swim lane, where each 'actor' has a lane of their own.
- Special forms and notations exist to denote tasks and activities:



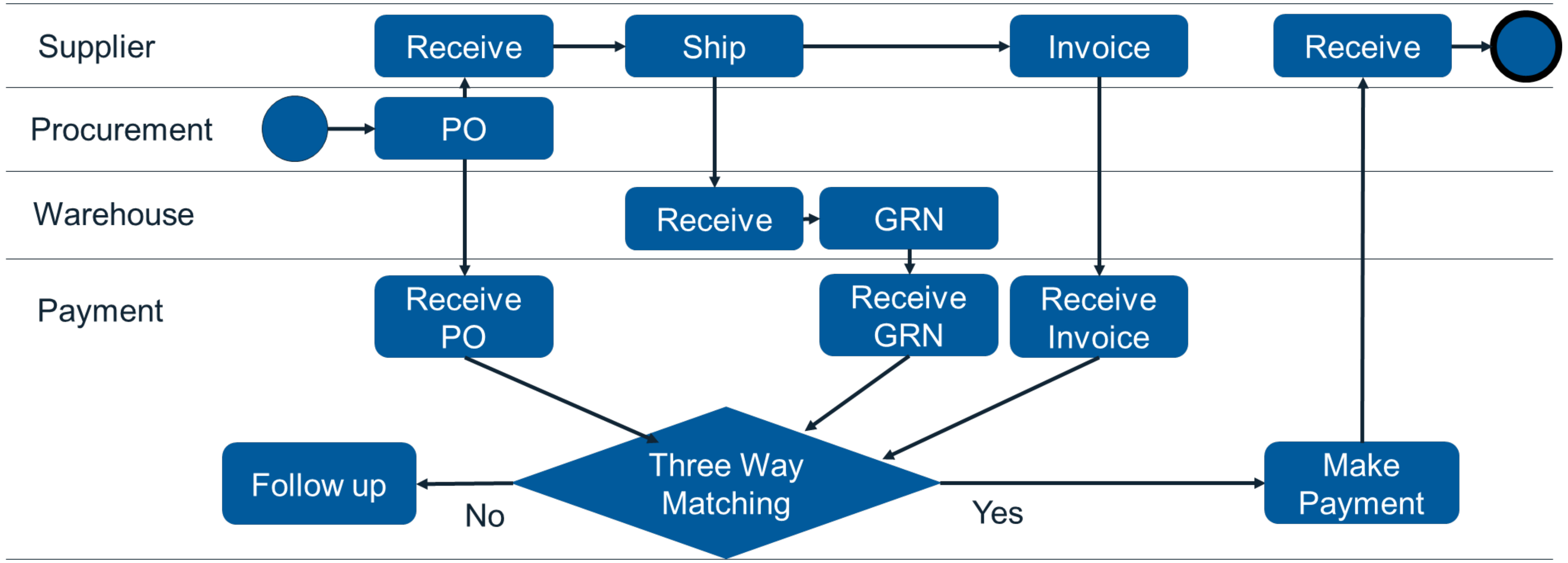
Example: Document Matching

- A simple ordering and payment process can be significantly streamlined through usage of IT (PO: purchase order, GRN: goods received note)



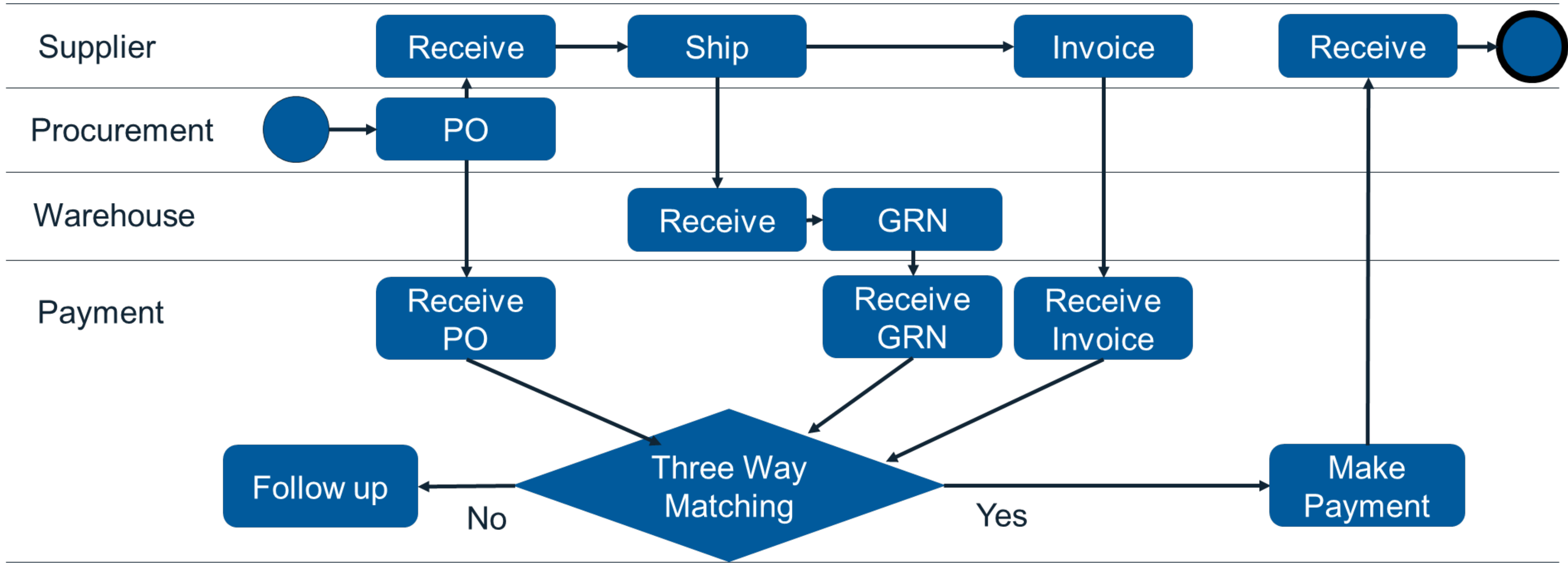
Why We Draw the Process

- The first step is drawing the present process to understand
  - Its purpose
  - Its inputs and outputs
  - Parties involved
  - Methods of information flow
  - Problem points



Objective: Streamlining and Automation

- Understand why we are doing it: reduce cost, reduce errors, reduce delay, etc
- (Generally, to sell to the management, although generally this should be a top-bottom approach…)
- List the technologies needed:
  - What are available technologies
  - What needs to be developed
  - Redraw flow, decision points



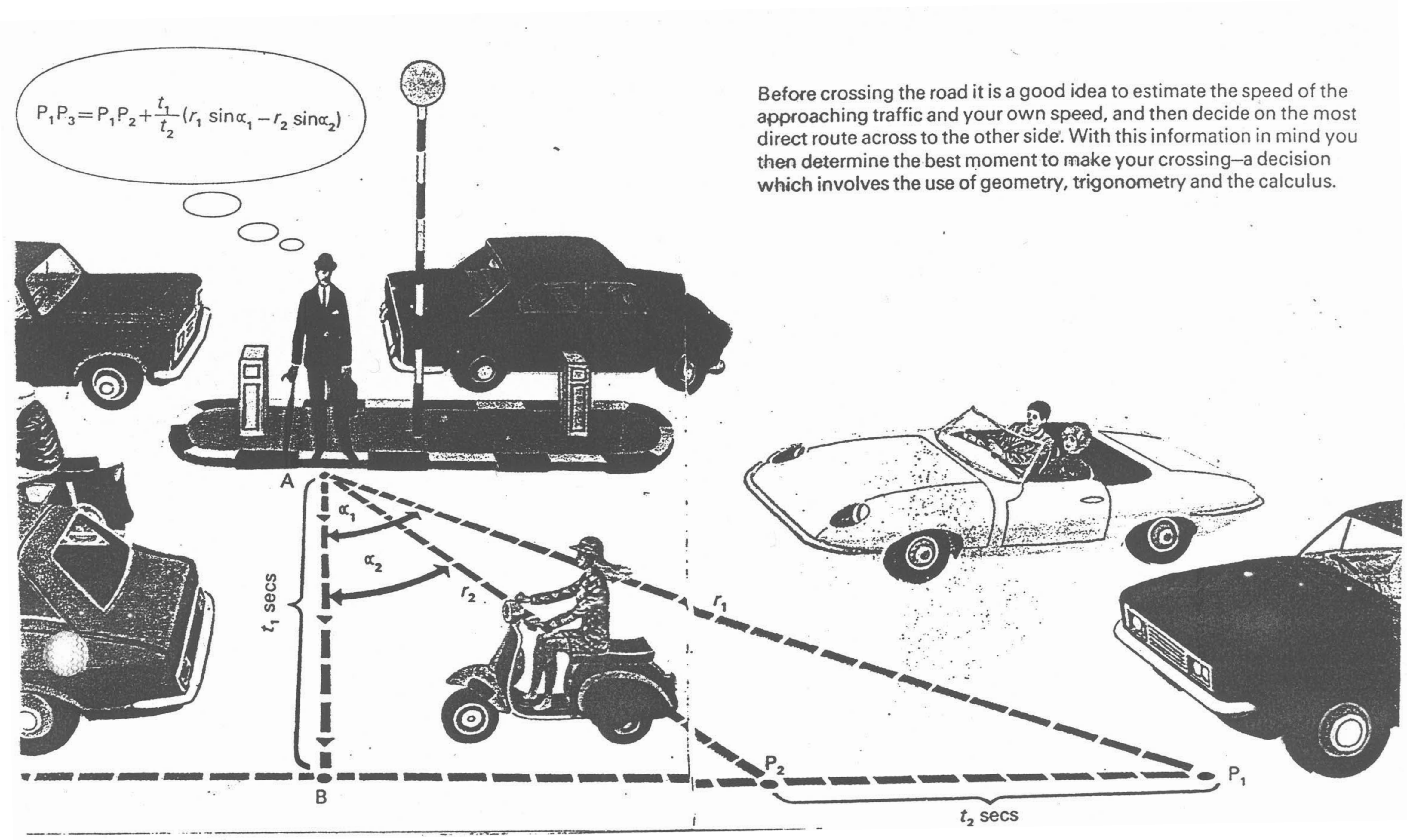
A Perfect Process

- A snake attack
- Snakes have 200-400 vertebrae whereas humans have approximately 33. Also, we have 640-850 muscles whereas snakes have some 4000. (eNotes, Kids Zone and Answers.com)
- Our walking is similarly complex…

An Almost Equally Perfect Process

- Some skills and some madness…
- René Higuita the national Columbian goal-keeper used this creative “scorpion” kick in a 1995 friendly match vs England

Crossing the Road



What Is Automated?

- Computers can replace some steps of a process
- Pre-requisite are:
  - These are repetitive
  - They are used often
- Machine Learning / Artificial Intelligence are methods where computer processing replaces a step of a process

Human Intelligence

- Throughout cycles we have learned what works and what doesn’t
- Part of these have come naturally – through a Darwinian process
  - More food, more population, more power over foe, more kids, more genes left to grow the clan, etc.
  - Better governance, happier population, more stability, more technological progress, more military power, empires, etc. (and of course the inevitable collapse…)
- One can see the beginning of growth of intelligence

Artificial Intelligence

- Replicated now in Artificial Intelligence as for example in first checkers program
  - A desired output is set against inputs and different way are tried to arrive at the final output

- The same learning process which partly powers AlphaGo.
- Note that much of human (at least early intelligence – but even to this day) is through a Darwinian process.
- Much the same way in the animal kingdom – as with our famous little yellow snake!

My Definition of Analytics

- Analytics is in essence finding correlation between sets of inputs and one or more output(s).
- Humans have been doing analytics, learning associations and correlations
  - The four seasons and growth and availability of food
  - Sounds and lack of sounds and association with food/danger/etc.
  - Fire with warmth and cooking and better diet and health (longer life)
  - Planting seeds and growing plants,

Intelligence in Social Networks

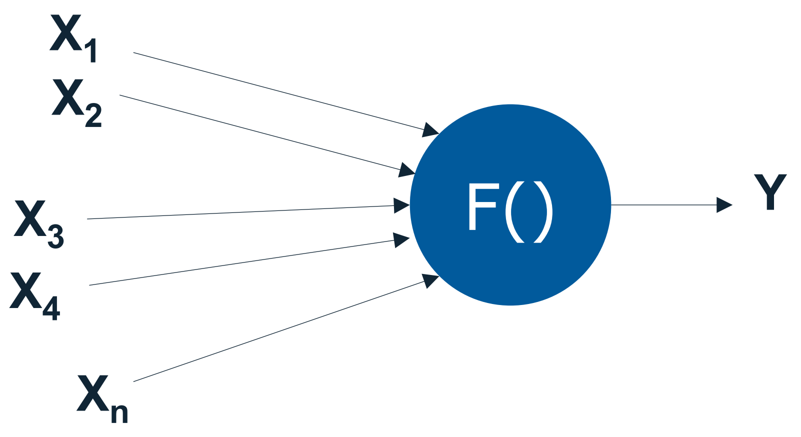
- We also see the same patterns of ‘analytics’ associated with trade, governance, politics, wars, ethics, etc.
- Basically what works and what doesn’t work…
- Note the difference between causality and correlation
- We are generally talking about correlation – although there usually is a degree of causation

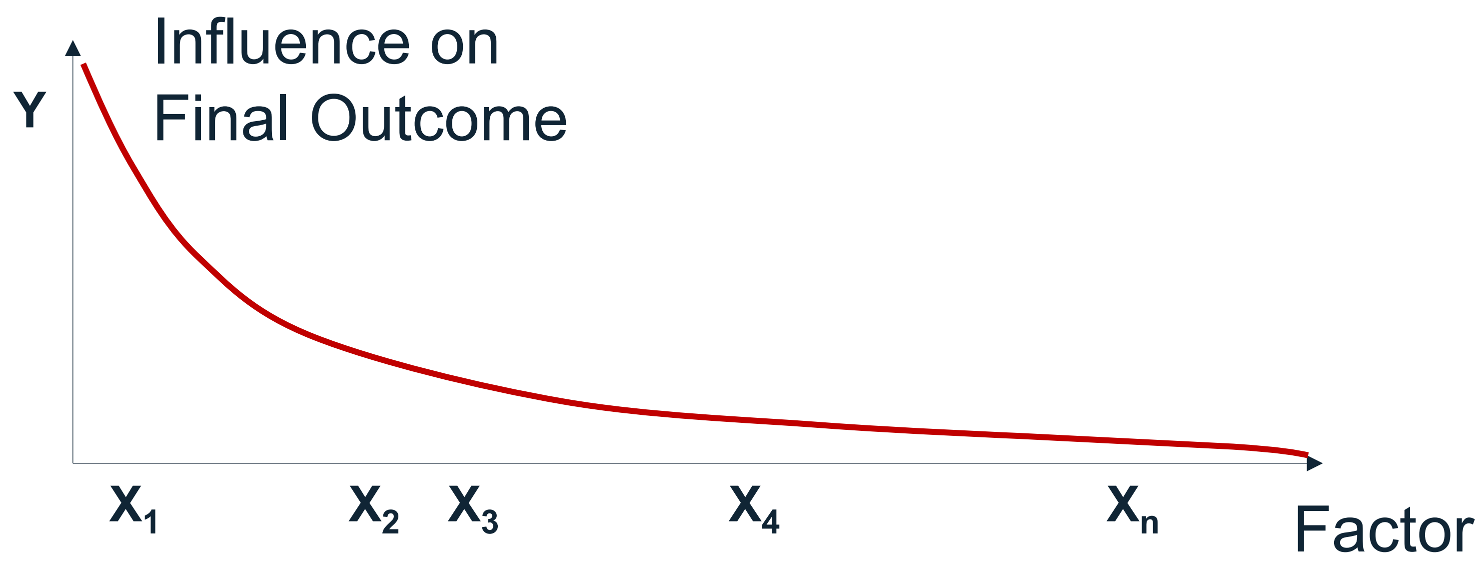
Further on Intelligence

- Clearly the flow of causation and correlation can be quite complex
- The factors that contribute can be also many, each with a varying degree of influence on the final outcome
  - Type of grape
  - Irrigation
  - Fertilization
  - Type of soil, Etc.

Formulating Correlation

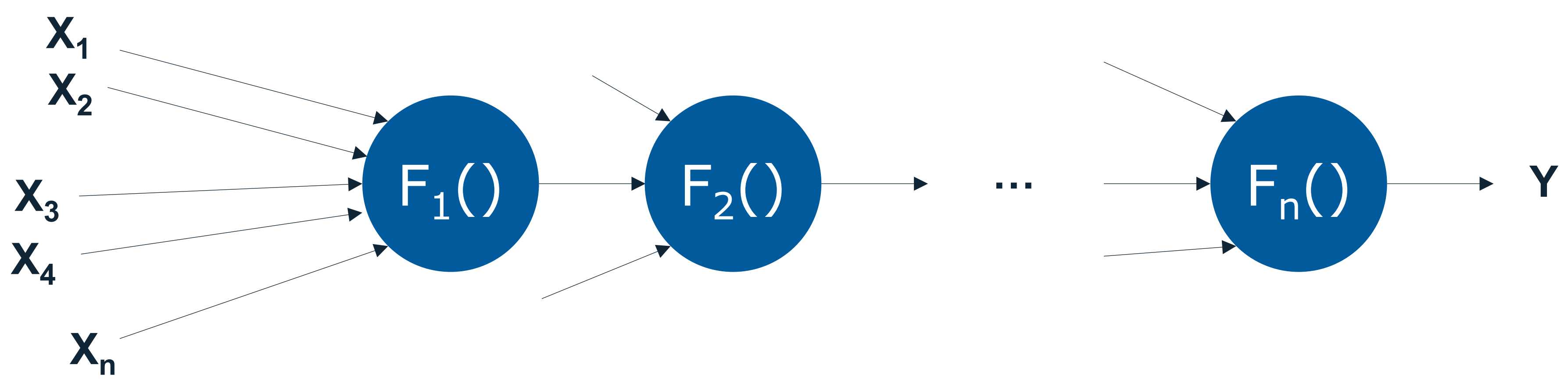
- In single-step processes, one can write formulas that connect input and output
  - **$Y = f(X_1, X_2, X_3, X_4, \ldots, X_n) + c$**





Multiple Step Correlations

- At times there exist intermediary steps before the final outcome
- Understanding the intermediary steps may provide opportunities for feedback loops and further intervention to enhance an outcome
- Learning about these correlations (or collecting and processing information on these systems) help enhance a desired outcomes



Summary

- Processes are more common than projects (by definition)
- As we repetitively carry out tasks, we become better at it
- Technology can be used/developed to redesigning, streamline processes and make them more efficient
- Digital transformation facilitates the automation of many steps within a process
- Artificial intelligence and machine learning also generally apply to processes